

When Does Graph Expansion Help Personal-Corpus Retrieval? A Query-Class-Stratified Evaluation over Human-Curated Wikilink Graphs: Protocol and First Execution with LLM-Judged Relevance (v0.3)

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Abstract

Graph-augmented retrieval (Graph RAG) systems usually build their graph by LLM-based entity and relation extraction. We study a setting that is common in personal knowledge management yet under-evaluated: the graph already exists as *human-curated wikilinks* in a linked note corpus. We describe a deployed retrieval system over such a corpus (multilingual-e5 dense retrieval, a bounded 1-hop wikilink expansion used strictly for candidate generation, cross-encoder reranking; no graph database, no extraction pass) and a query-class-stratified evaluation protocol (entity, theme, bridge, compare, temporal). Version 0.2 reported only $N=10$ pilot A/B pairs. This version reports the **first full execution** of the protocol: 38 queries, 509 pooled documents, graded 0–2 under blinding. **The relevance labels were produced by an LLM judge (Claude Opus 5.5), not by a human annotator**; to quantify how much that matters we re-graded a 14-query, 178-document subset with two further LLM judges from other vendors (GPT-5.6, GLM-5.3). Inter-judge agreement is moderate (Cohen’s κ 0.32–0.42 unweighted, 0.54–0.64 quadratic-weighted between the primary judge and the others; 0.57 / 0.73 between the two external judges). Under the primary labels, graph expansion is *inert or slightly harmful* on average: mean $\Delta\text{nDCG}@12 = -0.036$ over all 38 queries (12 harmed, 6 helped, 20 unchanged), with no stratum significant after Holm correction. The v0.2 hypothesis that expansion helps thematic queries is *not* supported (theme: $\Delta = -0.020$, 95% CI $[-0.041, -0.002]$, $n=16$); the sign of small per-query effects is judge-dependent. We report these as descriptive estimates from one corpus and one user, and we make the judge, not a human, explicit wherever a label is used.

1 Introduction

Retrieval-augmented generation over personal knowledge bases increasingly incorporates graph structure. The dominant recipe builds that structure automatically: an LLM extraction pass mines entities and relations, the triples populate a graph store, and retrieval traverses the result. This recipe carries three costs: an extraction pass that can hallucinate edges, a graph database to operate, and an ETL pipeline that can drift out of sync with the corpus.

A large population of corpora already contains a graph that no model produced: users of linked note-taking systems (Obsidian, Logseq, Roam, plain-Markdown wikis) place `[[wikilinks]]` by hand, over years. Each such edge is a deliberate, human-curated assertion that two notes are related. Whether retrieval over such corpora benefits from traversing these edges — and *for which*

classes of queries — is, to our knowledge, not systematically evaluated in the Graph RAG literature, which concentrates on automatically extracted graphs.

Contributions, in ascending order of importance:

1. **System description.** A minimal production retrieval stack over a human-curated wikilink corpus in which the graph is used *only* as candidate generation and is parsed at query time, never materialized (§3).
2. **Evaluation protocol.** A query-class-stratified, paired, blinded benchmark design with a pre-stated analysis plan (§4).
3. **First execution, with the label source made explicit.** Results on 38 queries with relevance graded by an LLM judge, plus a multi-vendor judge-agreement study that bounds how far those labels can be trusted (§5–§7). The headline is negative: on this corpus, a reranker-guarded 1-hop expansion does not measurably help any query class.

We claim no novelty for dense-plus-rerank retrieval, graph-assisted candidate generation, stratified evaluation, or LLM relevance judging as methods. The contribution is their combination in the human-curated-wikilink setting, and an honest first measurement.

2 Related work and honest prior art

- **Automatically extracted graph memory.** Microsoft GraphRAG builds an LLM-extracted entity graph with community summaries [1]. Graphiti/Zep maintain a temporally-aware knowledge graph [2]. HippoRAG runs Personalized PageRank over an automatically constructed KG [3]. All construct the graph by model extraction rather than consuming a human-curated one.
- **Agentic and tiered memory.** A-MEM lets the *agent* link notes Zettelkasten-style [4]; MemGPT/Letta manage tiered memory [5]. Neither evaluates retrieval over pre-existing human-placed links.
- **Retrieve-then-rerank.** Dense retrieval followed by cross-encoder reranking is standard [6]; we use it unmodified.
- **Stratified IR evaluation** has long precedent in TREC-style per-topic-type analysis [7].
- **LLM relevance judgments.** Using LLMs as relevance assessors has been studied directly, with agreement with human assessors reported as comparable to inter-human agreement on some collections and with documented failure modes elsewhere [8, 9]. We adopt the practice and, following that literature, report judge agreement rather than assume it.

3 System

3.1 Corpus

The deployment corpus is a single user’s personal knowledge base accumulated over about fifteen years: roughly 176,000 Markdown files, of which a curated slice of 4,591 files (concepts, decisions, insights, people and project cards) is indexed for semantic retrieval as 7,557 chunks. Raw chat archives, session logs and ledgers are excluded from the semantic index and served by lexical search. The corpus is mostly Russian with English technical terms.

3.2 Retrieval pipeline

1. **Entry points.** Dense retrieval with multilingual-e5 over the curated slice.
2. **Candidate generation by graph expansion.** For each top hit, the note’s wikilinks are parsed *at query time* and 1-hop neighbours are added to the pool, bounded at 40 per query. Edges receive no score and confer no bonus. For name-shaped queries a lexical predicate (`looks_like_entity()`) routes expansion through the entity’s card instead of the generic walk.
3. **Reranking.** A cross-encoder reranks the merged pool jointly; the top 12 are returned.

Two constraints define the system: expansion is candidate generation, not ranking (a graph-arrived note wins only if the reranker independently prefers it), and the graph is never materialized (no graph database, no extraction pass, no graph-corpus drift). Persistence is a single SQLite file.

4 Query-class-stratified evaluation protocol

4.1 Strata

Entity. The query names a person, tool, project or organization. Hypothesized *harm* regime (hub-node flooding).

Theme. The query names a topic without naming an entity. Hypothesized *help* regime.

Bridge. The answer lies in a note dissimilar to the query but linked from one that is similar. Hypothesized *strongest-help* regime.

Compare. The query asks for a contrast between two items.

Temporal. The query is anchored to a time or a sequence of decisions.

4.2 Conditions, pooling and blinding

Paired conditions per query: (A) vector-only; (B) vector + 1-hop expansion, both invoking the deployed retrieval program rather than a reimplementaion, and the deployed entity predicate rather than a copy. The top-12 of A and B are pooled. The labeling interface receives only title, date, a 200-character snippet and an opaque key, in an order shuffled by a per-query deterministic seed; which condition retrieved a document, at which rank, and whether it arrived through the graph stay in the database.

4.3 Metrics and analysis plan

Primary metric: paired $\Delta \text{nDCG}@12 = \text{nDCG}(B) - \text{nDCG}(A)$, gain $2^g - 1$. Secondary: pool-relative recall@12, *promotion rate* (share of queries where B surfaced a relevant note A missed), *harm rate* (share with $\Delta < -\varepsilon$, $\varepsilon = 0.01$) and *inert rate* ($|\Delta| \leq \varepsilon$). Per stratum: mean and median Δ , 10,000-resample bootstrap 95% CI, Wilcoxon signed-rank (exact for ≤ 18 non-zero differences), Holm correction across the five strata. Queries with an incompletely graded pool, no relevant document, or an empty condition are excluded and counted.

5 Execution (v0.3)

5.1 Query set

The run comprised 39 queries; one (an entity query) timed out after 900 s in retrieval and has no pool, leaving **38 queries and 509 pooled documents**. Provenance matters and we report it: 8 queries come from the production query log (all *theme*); 30 are authored seeds (entity 9, theme 8, bridge 5, compare 4, temporal 4). The log contains no recognizable bridge queries and almost no entity queries, so those strata are seed-only and therefore weaker evidence, as anticipated in v0.2. Of the 509 pooled documents, 36 reached condition B’s top-12 via the generic wikilink walk and 18 via the entity route; the rest were found by the vector lane.

5.2 Relevance labels: an LLM judge, not a human

All relevance labels used for the primary results were produced by an LLM judge (Claude Opus 5.5, model id claude-opus-5-5), not by a human annotator. The protocol as written in v0.2 called for grading by the corpus owner; that step was not performed, and we do not present these labels as a substitute for it. The judge saw exactly what a human labeler would see: the query and, for each pooled document, title, date and the 200-character snippet, in the blinded shuffled order, with no access to condition, rank or graph provenance. It graded every document 0 (not relevant), 1 (partially relevant: related topic or useful context) or 2 (directly answers or is centrally about the query). Of 509 documents it graded 204 as 0, 175 as 1 and 130 as 2; every one of the 38 pools contained at least one relevant document. The judge is the same model family that operates parts of the system’s surrounding tooling; it did not build the retrieval layer under test, but this proximity is a threat we list in §8.

5.3 Judge-agreement study

To estimate label reliability we drew a subset of 14 queries (178 documents; at least two per stratum; queries whose snippets contained personal contact data were excluded for privacy) and had it graded independently, with the same instructions and the same blinded view, by two LLMs from other vendors: GPT-5.6 (OpenAI) and GLM-5.3 (Zhipu). A third external judge (Mistral) returned outputs with wrong lengths and near-constant grades and was discarded as invalid; a fourth (DeepSeek) exhausted its output budget on reasoning and returned no grades. GLM-5.3 returned malformed grade strings for 2 of 14 queries, which are excluded from its comparisons. Table 1 reports agreement.

pair	docs	exact	κ	κ_{quad}	κ (rel $\geq$ 1)	κ (rel=2)
Claude – GPT-5.6	178	0.573	0.321	0.541	0.410	0.429
Claude – GLM-5.3	152	0.618	0.418	0.644	0.710	0.411
GPT-5.6 – GLM-5.3	152	0.750	0.570	0.729	0.531	0.695
GLM-5.3 run 1 – run 2	125	0.760	0.608	0.765	0.604	0.728

Table 1: Inter-judge agreement on the 14-query subset (Cohen’s κ ; quadratic-weighted for the 3-point scale; binary at two relevance thresholds). The last row is the same model queried twice and bounds self-consistency.

Agreement is moderate. The primary judge is systematically *stricter* than both external judges (on the subset it assigned grade 2 to 35% of documents, versus 57% for GPT-5.6 and 54% for

GLM-5.3, and grade 0 to 22% versus 7% and 18%). The two external judges agree with each other more than either agrees with the primary judge, and GLM-5.3 agrees with itself across two runs only at $\kappa = 0.61$. These figures are the honest reliability ceiling for every number below.

6 Results

stratum	n	mean Δ	median Δ	95% CI	p (Holm)	promo / harm / inert	recall A $\rightarrow$ B
entity	9	-0.057	0.000	$[-0.129, 0.000]$	1.00	11 / 33 / 67%	0.99 $\rightarrow$ 0.97
theme	16	-0.020	-0.004	$[-0.041, -0.002]$	0.34	19 / 38 / 50%	0.98 $\rightarrow$ 0.92
bridge	5	-0.082	0.000	$[-0.261, 0.017]$	1.00	20 / 20 / 60%	0.99 $\rightarrow$ 0.79
compare	4	-0.036	+0.079	$[-0.233, 0.097]$	1.00	75 / 25 / 25%	0.85 $\rightarrow$ 0.77
temporal	4	+0.002	0.000	$[-0.042, 0.048]$	1.00	50 / 25 / 50%	0.95 $\rightarrow$ 0.95
all	38	-0.036	0.000			12 harmed / 6 helped / 20 inert	

Table 2: Condition B (vector + 1-hop wikilink) versus A (vector only), primary LLM-judge labels. $\Delta = \text{nDCG@12(B)} - \text{nDCG@12(A)}$. Raw p before correction: theme 0.067, entity 0.25, others 1.0. Recall is pool-relative. Promotion counts queries where B surfaced a relevant note A missed and can co-occur with harm.

Mean nDCG@12 is 0.778 for A and 0.742 for B. No stratum shows a significant effect after Holm correction, and all strata are far below the 30-query size the protocol names as the minimum for confirmatory testing, so the table is descriptive. Three observations nonetheless bear on the v0.2 hypotheses.

Theme queries are not helped. v0.2 conjectured, from inspection of 10 pilot pairs, that thematic queries benefit. With 16 theme queries the point estimate is negative and its bootstrap interval excludes zero by a hair; the Wilcoxon test is not significant ($p_{\text{raw}} = 0.067$). The defensible reading is “no benefit detected,” not “harm demonstrated.”

Entity harm is concentrated. Two thirds of entity queries are inert; the negative mean comes from 3 of 9 queries in which the entity route displaced directly relevant notes. This matches the hub-flooding mechanism qualitatively but cannot establish it at $n = 9$.

The graph mostly does nothing, which is what the design intends. Over all 38 queries, 20 are inert; across the whole run the graph contributed 11 relevant promotions. A reranker-guarded expansion should be inert often; the operational question is whether its occasional promotions outweigh its occasional demotions, and on this corpus, under these labels, they do not.

7 Sensitivity to the judge

Because labels come from an LLM, we re-scored the 14-query subset under each judge separately (Table 3).

labels	entity	theme	bridge	compare	temporal
Claude (subset)	-0.028 (3)	+0.004 (4)	+0.009 (3)	+0.040 (2)	+0.003 (2)
GPT-5.6	-0.004 (3)	-0.011 (4)	-0.021 (3)	+0.004 (2)	+0.003 (2)
GLM-5.3	0.000 (2)	+0.004 (3)	+0.005 (3)	+0.034 (2)	+0.027 (2)

Table 3: Mean $\Delta \text{nDCG@12}$ per stratum on the agreement subset under each judge (number of queries in parentheses).

All subset effects are within ± 0.04 , and their signs differ between judges in three of five strata. Classifying each query’s Δ as harmed / inert / helped ($\varepsilon = 0.01$), the primary judge agrees with GPT-5.6 on 8 of 14 queries (one opposite sign) and with GLM-5.3 on 11 of 12 (none opposite); the two external judges agree on 7 of 12 (one opposite). We conclude that on this corpus the effect of graph expansion is small enough that its sign is not robust to the choice of LLM judge. Note also that the subset happens to be more favourable to B than the full set under the primary judge, a reminder that 14 queries do not represent 38.

8 Threats to validity

- **LLM labels, not human labels.** No human relevance judgment enters any number in this paper. LLM judges can share the biases of the retrieval models they evaluate, and agreement among LLMs is not agreement with the corpus owner, who alone knows what is relevant on a personal corpus. Human grading of at least the agreement subset is the most important next step.
- **Judge proximity.** The primary judge belongs to the model family that runs surrounding tooling in the same deployment. It saw no condition information, but it was not independent of the deployment’s authorship.
- **Snippet-only view.** All judges saw 200-character snippets, as a human labeler would in this interface. Relevance that only the full note reveals is invisible to every judge equally.
- **Seed-heavy strata.** Entity, bridge, compare and temporal strata are entirely author-written seeds; whoever writes seeds can flatter or handicap the graph without noticing.
- **Single subject, single corpus, small strata.** One user, one curation style, 4–16 queries per stratum.
- **Pool-relative recall.** Unretrieved documents are never graded, so recall is relative to the union of the two conditions.
- **Stratification construct validity.** Entity membership is decided by the production predicate; other strata by the seed author. No second annotator classified strata in this run.
- **Ablations not run.** Hop depth, neighbour cap, entity gate and pool size were specified in v0.2 but not executed here.

9 Replicability without data disclosure

The corpus is private and cannot be released. The protocol is portable rather than reproducible on our data: the retrieval layer is public (MIT), and the four standard-library programs that pool candidates, build the blinded interface, ingest grades and compute the analysis ship with stratification rules and judge instructions. Any owner of a linked Markdown corpus can run the identical protocol, with a human or an LLM judge, and report agreement as we do here.

10 Conclusion

We executed a query-class-stratified, paired, blinded evaluation of reranker-guarded wikilink expansion on a personal corpus for the first time. With relevance graded by an LLM judge and that

judge’s reliability bounded by two external judges (κ 0.32–0.42), graph expansion is inert on half of all queries and slightly negative on average, and no query class shows a significant benefit; the effects are small enough that their sign depends on the judge. The v0.2 hypothesis that thematic queries benefit is not supported. Next steps are human grading of the agreement subset, enlarging each stratum to the protocol’s 30-query minimum, and the ablation grid.

Data and code availability

Code for the retrieval layer and the evaluation programs is public via <https://github.com/tonydz>. Per-query results, label files and the judge prompt are retained privately because pooled snippets come from a private corpus; aggregate tables in this paper are complete. Contact: dzyatkovskiy.a2@gmail.com, ORCID 0000-0001-7408-3054, <https://tonydz.github.io>.

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